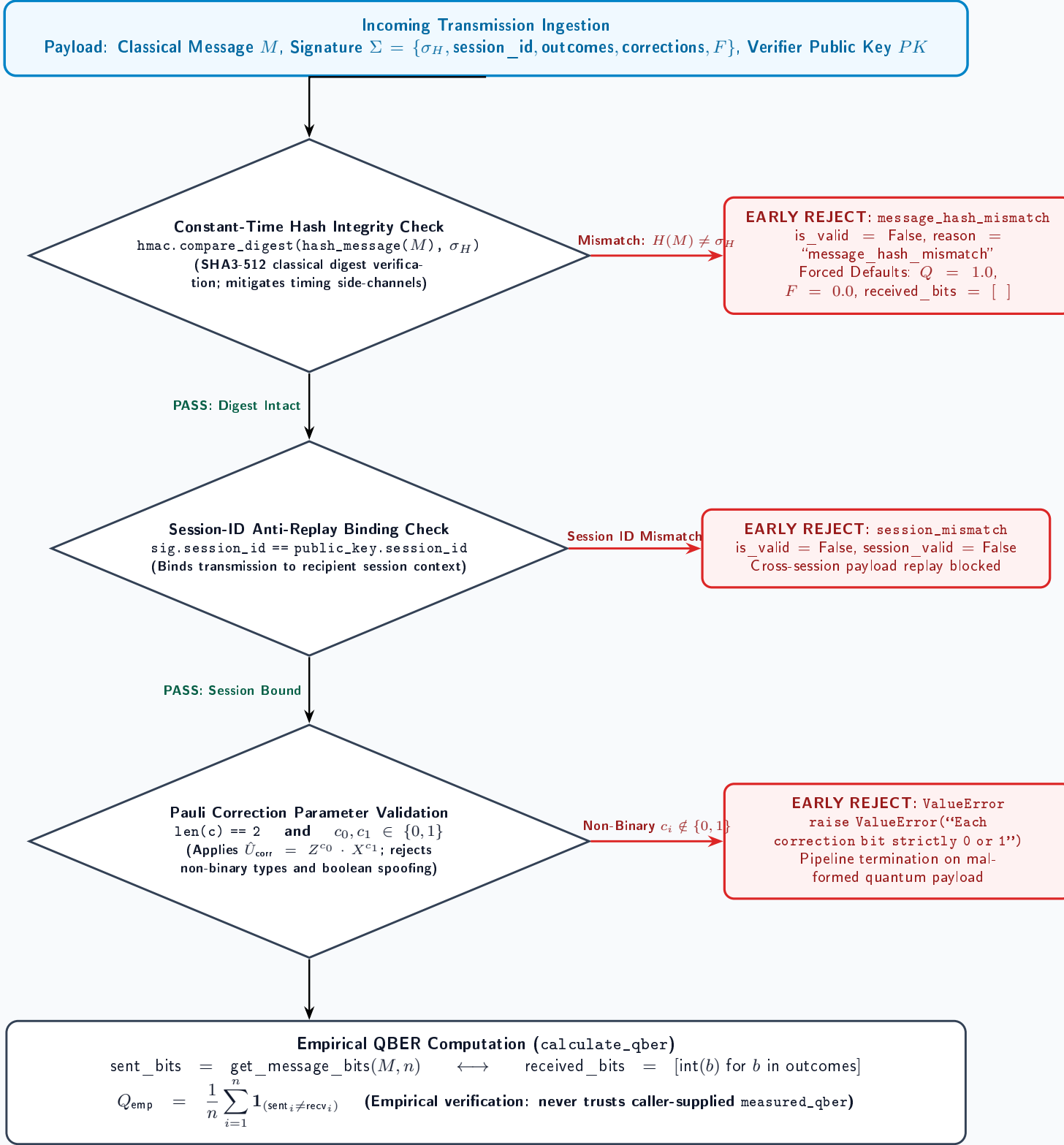


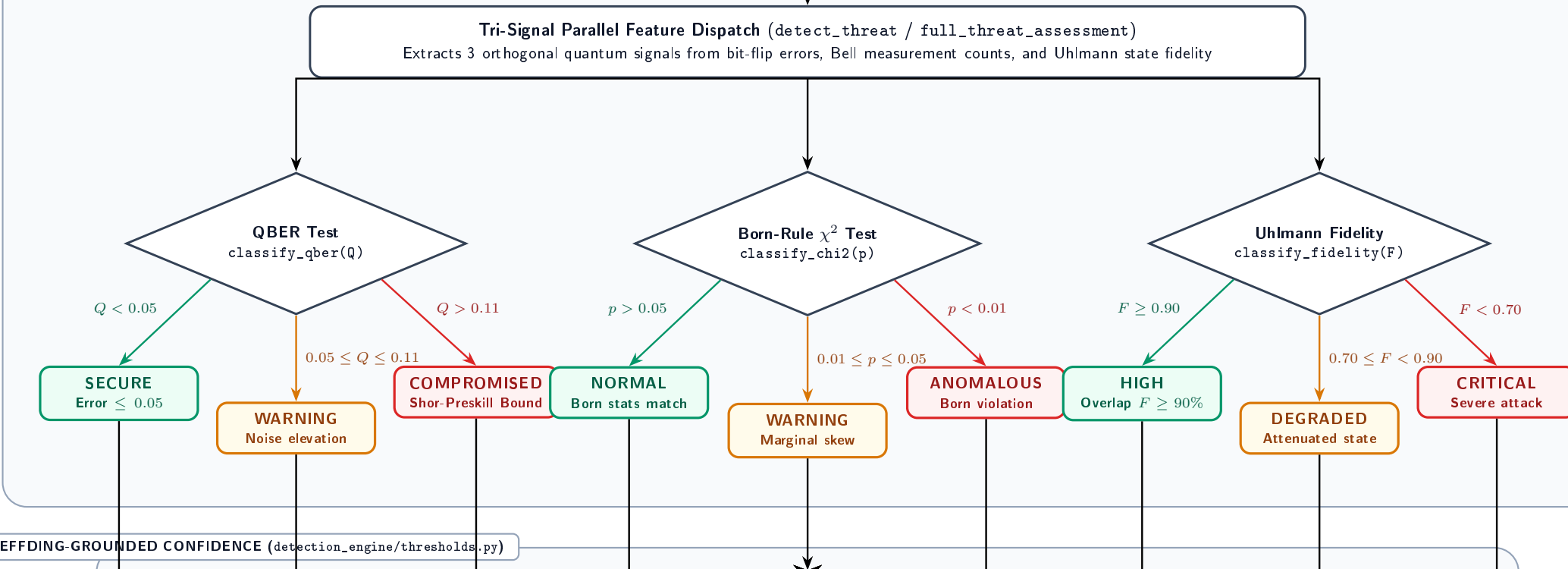
Actual QDS Threat Detection & Verification Pipeline

Grounded Decision Logic: Classical Pre-Gateways → Tri-Signal Quantum Statistics → Hoeffding Threat Confidence → SHA3-512 Hash-Chained Audit Ledger

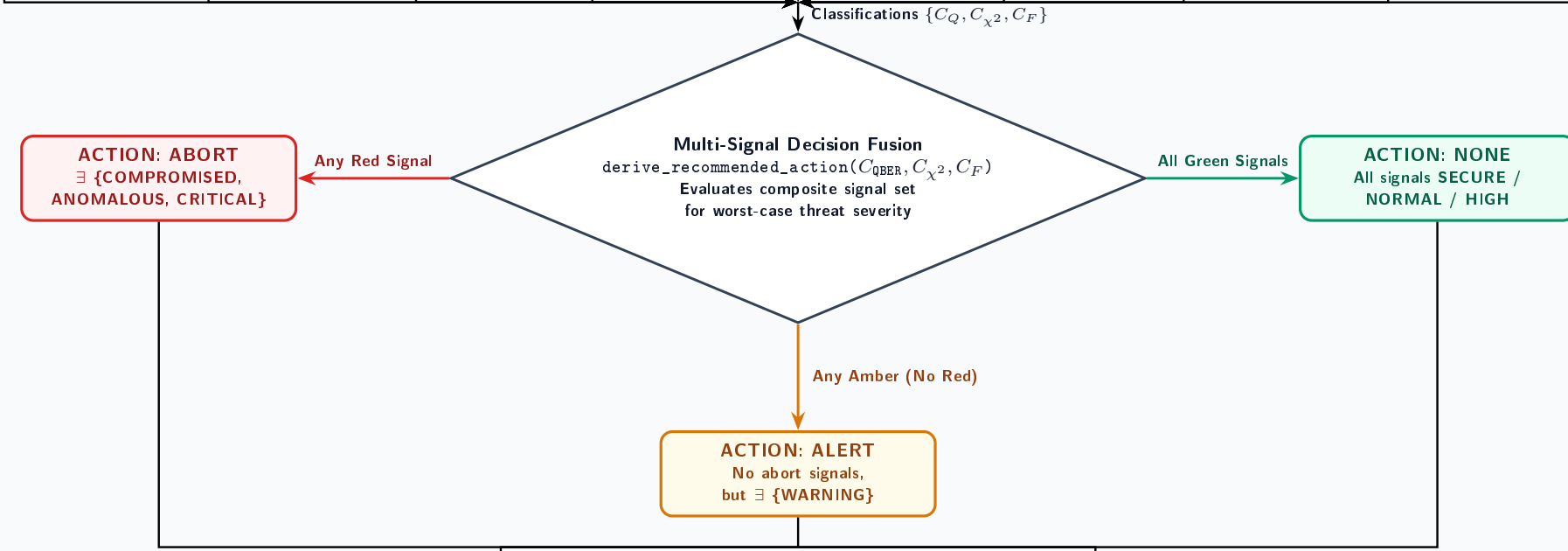
PHASE 1: UPSTREAM CLASSICAL & QUANTUM GATEWAYS (qds_core/verification.py)



PHASE 2: TRI-SIGNAL PARALLEL QUANTUM STATISTICAL CLASSIFICATION (detection_engine/detector.py)



PHASE 3: DECISION FUSION & Hoeffding-GROUNDED CONFIDENCE (detection_engine/thresholds.py)



PHASE 4: DETERMINISTIC VERDICT & POST-QUANTUM CRYPTOGRAPHIC PERSISTENCE (backend/audit_ledger.py)

